

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

Sunday 14th June, 2026

 Hong Kong
 spatlyu.github.io
 lyu.geosocial@gmail.com
 SpatLyu
 0009-0002-6003-3800

Some stuff about me

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.

Education

2021.8-2025.6 **B.Sc. In Geographic Information Science**
Xi'an, Shaanxi

Shaanxi Normal University

Research Experience

2026.6-2026.11 **Research Assistant**
Hong Kong

The Hong Kong Polytechnic University

2024.8-2026.6 **Research Intern**
Guangzhou, Guangdong

The Hong Kong University of Science and Technology (Guangzhou)

Publications

1. Lyu, W., Dai, S., Song, Y., Zhao, W., Yi, W., Xiao, Y., & Jia, N. (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*. <https://doi.org/10.1080/13658816.2026.2687121>
2. Lyu, W., Lei, Y., Yi, W., Song, Y., Li, X., Dai, S., Qin, Y., & Zhao, W. (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems*, 127, 102435. <https://doi.org/10.1016/j.compenvurbsys.2026.102435>
3. Lv, W., Lei, Y., Liu, F., Yan, J., Song, Y., & Zhao, W. (2025). gdverse: An R package for spatial stratified heterogeneity family. *Transactions in GIS*, 29(2), e70032. <https://doi.org/10.1111/tgis.70032>
4. Lv, W., Liu, F., Cai, K., Cao, Y., Deng, M., Liang, W., Yan, J., & Wang, G. (2024). Distinguishing the impacts and gradient effects of climate change and human activities on vegetation cover in the weihe river basin, china. *Journal of Geophysical Research: Biogeosciences*, 129(10), e2024JG008297. <https://doi.org/10.1029/2024jg008297>
5. Chen, C., Song, Y., Lv, W., Shemery, A., Hampson, K., Yi, W., Zhong, Y., & Wu, P. (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. <https://doi.org/10.1111/mice.13489>
6. Xiao, Y., Yang, Y., Zhao, T., Wang, W., Lv, W., & Zhao, W. (2026). Contrasting causal pathways of vegetation greening between economically strong and weak towns in china's greater bay area. *GIScience & Remote Sensing*, 63(1), 2623327. <https://doi.org/10.1080/15481603.2026.2623327>

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.

Unpublished

Co-First Author (1st)	Quantifying pairwise directional influence between spatial variables: Asymmetry, nonlinearity, and beyond	Submitted to AAAG, currently under review
Co-First Author (2nd)	Spatial Pattern Entropy: A Symbolic Measure for Continuous Spatial Data	Submitted to IJGIS, currently under review
First Author	Unveiling interaction between spatial processes with information decomposition	In writing

First Author	Pattern Causality for Detecting Interactions with the R Package pc	In writing
First Author	coupling: Coupling Coordination Analysis in R	Plan
First Author	Spatial causal discovery under latent confounders via deep orthogonal representation learning	Plan

Developed Spatial Analysis Toolkit

Package	Description	Language
spEDM	Spatial Empirical Dynamic Modeling	C++, R
tEDM	Temporal Empirical Dynamic Modeling	C++, R
infoxtr	Information-Theoretic Measures for Revealing Variable Interactions	C++, R
pc	Pattern Causality Analysis	C++, R
gdverse	Analysis of Spatial Stratified Heterogeneity	R, C++, Python
sdsfun	Spatial Data Science Complementary Features	R, C++
coupling	Coupling Coordination Analysis	C++, R
gobi	General ODE-Based Causal Inference	C++, R
geocomplexity	Mitigate Spatial Bias Through Geographical Complexity	C++, R, C
HSAR	Hierarchical Spatial Autoregressive Model	C++, R
GD	Geographical Detectors for Assessing Spatial Factors	R
geosimilarity	Geographically Optimal Similarity	R